QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 10

Climate Friendly Movements

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	10.0 - Climate Friendly Movements (15 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Mapping rule	Map only what the lecture actually shows. The report preserves the deck's own figures and numerical claims, and labels a question Partial when the deck merely names a topic (for example, a full traditional-vs-contemporary material comparison or a detailed modern Tube House reinterpretation).

RESULT	Lecture 10 is a high-return climate-friendly construction lecture. Rat-Trap Bond repeats in four consecutive papers (2020-2023), Tube House appears in 2016, 2021, 2022 and 2024, and Green/Sustainable/Organic topics recur across several years. The best-return slide blocks are 3-4, 7-13 and 14-15.

Prepared from the uploaded lecture deck and the supplied 2016-2024 regular final question bank.

1. Executive Summary

Lecture 10 collects climate-friendly architectural and construction approaches: Tube House, Organic Architecture, climate-responsive design, sustainable/green building ideas, contemporary materials and Rat-Trap Bond construction. The deck is strongest where it provides drawings and construction details - especially Tube House and Rat-Trap Bond. It is weaker where it only names broad topics such as contemporary materials and technology.

15	17	9	2	6	147
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Highest-priority slide blocks

Slides	Lecture block	Historical signals	Exam value
3-4	Tube House: plan/section, day-night spatial use, ventilation	2016; 2021; 2022; 2024	Very high. Sketch-driven short/medium question with repeated wording.
5-6	Organic architecture + aerodynamic form/cross ventilation	2016; supports climate-responsive answers	Good short-note block; concept is present, but no long formal definition list.
7-13	Rat-Trap Bond: definition, construction, junction/opening, pointers, advantages	2020-2023	Highest-value construction block; direct repeated question family.
14-15	Sustainable vs Green + Green Building criteria	2017; 2018; 2023-2024 applications	Strong for definitions/comparison; applied sustainability cases need broader reasoning.
2	Movement overview + contemporary materials/technology	2016; 2017; 2020	Useful index only; full material comparison is not developed.

Priority note	Prioritize Rat-Trap Bond -> Tube House -> Green vs Sustainable -> Organic Architecture. Treat “Contemporary materials and technology” as a coverage gap because the lecture lists the topic but does not supply a complete climatic comparison.

2. Detailed Year-wise Question Mapping

The original printed section label may change from year to year. Mapping below follows subject matter and the lecture content. Slide numbering uses the PDF page count, including the title slide.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Q8(b-i)	part of 8	1	Explain Tube House with example.	Exact	3-4	The deck provides Tube House plan/section and day/night spatial use with natural-ventilation logic.
2016	Q8(b-ii)	part of 8	1	Explain Organic Architecture with example.	Direct	5-6	Nature-responsive form, cross ventilation, insulation, topsoil/trees/softscape and aerodynamic form are shown.
2016	Q8(b-iii)	part of 8	1	Explain Green Building Movement with example.	Direct	14-15	The deck defines green-building performance ideas and shows measurable criteria, though movement history is not developed.
2016	Q8(b-iv)	part of 8	1	Explain contemporary building materials.	Partial	2, 5	“Contemporary materials and technology” is named, and some material examples appear, but there is no complete climatic comparison.
2017	Q8(a)	10	2	Define Green Building Movement.	Exact	14-15	Direct green-building definition/criteria block.
2017	Q8(b)	25	2	Evaluate and compare contemporary and traditional building materials from the climatic point of view.	Partial	2, 5	The deck names contemporary materials and gives isolated material strategies; a full comparison table is not supplied.
2018	Q1(c)	10	3	Distinguish between sustainable building and green building.	Exact	14-15	Slide 14 directly contrasts the two ideas; slide 15 supports green-building criteria.
2020	Q2(a)	18	5	Show the construction technique of a brick wall using Rat-Trap Bond.	Exact	7-13	Definition, brick orientation, cavity, construction rules, opening/junction details and advantages are present.
2020	Q5(b)	15	5	Evaluate and compare contemporary and traditional building materials from climatic viewpoint.	Partial	2, 5	Lecture 10 does not provide a full traditional-vs-contemporary comparison.

2. Detailed Year-wise Question Mapping - continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2021	Q7(c)	7	7	Show the working mechanism of Tube House air ventilation.	Exact	3-4	Tube House sections show inlet/outlet airflow and day/night spatial logic.
2021	Q7(d)	8	7	Illustrate the working mechanism of Rat-Trap Bond in brick bonding.	Exact	7-13	The cavity-forming bond and detailing are explicitly illustrated.
2022	Q7(b)	8	9	Show the working mechanism of Rat-Trap Bond in brick bonding.	Exact	7-13	Direct repeat of the lecture construction block.
2022	Q7(c)	7	9	Show the working mechanism of Tube House air ventilation.	Exact	3-4	Direct repeat of the Tube House drawings.
2023	Q8(a)	10	12	Analyze sustainability of the KUET academic building under heat-wave, light/air/social, cost and maintenance conditions.	Partial	14-15	The sustainability/green framework is present, but the case-analysis dimensions are not fully worked in the lecture.
2023	Q8(c)	8	12	Illustrate the working mechanism of Rat-Trap Bond in brick bonding.	Exact	7-13	Direct repeated construction question.
2024	Q8(c)	6	15	Explain Tube House concept and spatial/climatic characteristics; discuss modern urban reinterpretation.	Partial	3-4	Original concept and spatial/ventilation logic are present; modern urban reinterpretation is not explicitly taught.
2024	Q8(d)	7	16	Analyze sustainability of the KUET academic building under heat-wave and user-performance constraints.	Partial	14-15	Framework support is present; the applied case requires additional synthesis.

Coverage key	Exact = answer content is essentially present as asked. Direct = most content is present, but the student must apply/combine it. Partial = only one component is present; another lecture/source is required.

3. Repeated Question Patterns

The lecture does not generate one single question type; it produces four recognizable families. Two of them - Rat-Trap Bond and Tube House - are repeated with very stable wording and are therefore the safest preparation targets.

P	Normalized repeated family	Years seen	Slides	Why it matters
S	Rat-Trap Bond: construction / working mechanism / thermal advantage	2020, 2021, 2022, 2023	7-13	Four consecutive papers; deck gives the exact construction sequence and diagrams.
S	Tube House: concept / ventilation mechanism / spatial-climatic value	2016, 2021, 2022, 2024	3-4	Repeated across four papers; easy sketch-based marks.
A	Green / Sustainable / Organic climate-friendly concepts	2016, 2017, 2018, 2023, 2024	5-6, 14-15	Common short/comparison/application cluster.
B	Traditional vs contemporary materials from climatic viewpoint	2017, 2020 (+ 2016 short)	2, 5	High marks but lecture coverage is incomplete; prepare from another source.

4. Quick Recognition Rules

If the question says...	Go to slides	Recognition rule
“Rat-Trap Bond / working mechanism / construction”	7-13	Draw the cavity bond, then write brick orientation -> cavity -> solid courses/opening details -> advantages.
“Tube House / air ventilation”	3-4	Use the section: low fresh-air entry, internal spatial flow, high warm-air exit; label day/night use if relevant.
“Green vs Sustainable”	14-15	Green = measurable environmental/health improvements; Sustainable = broader environmental + social + economic balance.
“Organic Architecture”	5-6	Nature-responsive form + cross ventilation + insulation + landscape/topsoil + aerodynamic form.
“Contemporary vs traditional materials”	2, 5	Do not overclaim: the deck only gives partial material strategies; another source is needed for a full comparison.

5. Slide-to-Question Reverse Index

Use this page while revising the lecture: start from a slide block, then immediately practice the linked past questions.

Slides	What the slides teach	Past-question links	Study action
2	Six climate-friendly movement headings	2016 Q8(b); materials questions support	Use as a lecture map, not a complete answer.
3-4	Tube House plan, sections, ventilation, day/night spatial adaptation	2016 Q8(b-i); 2021 Q7(c); 2022 Q7(c); 2024 Q8(c)	Practice one clean section with airflow arrows and labels.
5-6	Organic homes: form, cross ventilation, insulation, softscape, aerodynamics	2016 Q8(b-ii)	Prepare a 5-8 mark short note; identify the nature/climate response.
7-10	Rat-Trap Bond definition, cavity, brick orientation, construction/junction	2020 Q2(a); 2021 Q7(d); 2022 Q7(b); 2023 Q8(c)	Must-practice cavity sketch and construction sequence.
11-13	Rat-Trap Bond pointers, openings, reinforcement/services, advantages	Same Rat-Trap family	Memorize detailing rules and advantages; preserve slide-specific values.
14-15	Sustainable vs Green + Green Building criteria	2017 Q8(a); 2018 Q1(c); 2023/2024 sustainability cases partial	Prepare comparison table + green criteria list.

6. Coverage Gaps & Diagnostic Exclusions

Lecture 10 is strong for illustrated examples but uneven in depth. Broad labels on Slide 2 should not be expanded as if the deck contains a complete theory chapter.

Gap	Affected questions	Action
Full traditional vs contemporary material comparison	2017 Q8(b); 2020 Q5(b)	Use class/reference material; the lecture only names the topic and gives scattered material examples.
Modern urban reinterpretation of Tube House	2024 Q8(c)	Use the original Tube House principles as base, but mark the modern reinterpretation as outside this deck.
Applied sustainability audit of KUET academic building	2023 Q8(a); 2024 Q8(d)	Slide 14 gives the broad sustainability framework; detailed case evaluation requires additional reasoning/source.
History/evolution of the Green Building Movement	Any history-focused question	Not developed in the deck; do not invent a chronology from Slide 15.

Mapping discipline	The deck contains two Rat-Trap saving statements in different slides: Slide 9 says approximately 30% material is saved, while Slide 13 says approximately 25% fewer bricks and 40% less mortar. This report preserves both statements as slide-specific claims instead of silently reconciling them.

7. Recommended Study Plan

Study the lecture by return-on-time rather than slide order. First lock the two sketch-driven repeaters, then finish the sustainability vocabulary and finally identify the topics that need another source.

Step	Study block	Slides	Practice immediately
1	Rat-Trap Bond construction + advantages	7-13	2020 Q2(a), then 2021/22/23 working-mechanism variants.
2	Tube House mechanism + spatial logic	3-4	2021 Q7(c), 2022 Q7(c), then 2024 Q8(c) concept portion.
3	Green vs Sustainable + Green criteria	14-15	2017 Q8(a), 2018 Q1(c).
4	Organic Architecture	5-6	2016 Q8(b-ii) short note.
5	Coverage gaps	2, 5 + other source	2017/2020 material comparison; 2023/2024 applied sustainability.

Exam-ready minimum

Priority	What to reproduce	Exam technique
A	Rat-Trap cavity wall sketch	Label cavity, solid courses/opening zones, reinforcement/service possibility.
A	Tube House section	Show fresh-air path, internal space/court, high warm-air outlet.
A	Green vs Sustainable table	State broader social-economic-environmental sustainability vs green performance focus.
B	Organic architecture bullet cluster	Form + cross ventilation + insulation + landscape + softscape/aerodynamics.

Bottom line	Lecture 10 is worth mastering because its two strongest topics are predictable and sketchable. Rat-Trap Bond plus Tube House alone cover a large fraction of the lecture's repeated question history; Green/Sustainable concepts are the next-best short-note backup.